

LOAN PORTFOLIO RISK ANALYTICS

Loan Default Prediction

Executive summary & model prediction

13th July 2026

Home Equity Loan Portfolio (HMEQ) • 5,960 Applicants Analyzed



The Business Problem

The Business Problem Statement

Banks and Credit Unions process about 30% of residential mortgages and hold roughly 65% of outstanding home equity lines of credit. Mortgage and home equity lending typically generates between 10% to 30% of a retail bank's total net revenue. And the overall mortgage delinquency rate is approximately 4.44% and the default is at ~1.25% (nationwide).

Every loan approval is a bet on repayment.

- The studies from the report show that roughly 1 in 5 applicants in our portfolio ultimately defaulted and today's process relies heavily on manual judgment and static rules.

Leading to two costly failures:

- Missed risk: high-risk applicants get approved and later default, driving credit losses.
- Missed opportunity: creditworthy applicants get unnecessary friction or delay from overly cautious, one-size-fits-all review.

The Objective

Project Objective

- Build a data-driven model that predicts default risk at the time of loan application.
- Offer recommendations to the bank on the important features and process improvements to consider while approving a loan.

The Dataset

THE DATASET KEY OBSERVATIONS

5,960

home equity loan applications, each with 13 financial and demographic attributes

19.95%

~ 1190 of 5960 applicants in the historical data defaulted, a meaningful number of defaulters that the model must learn to spot

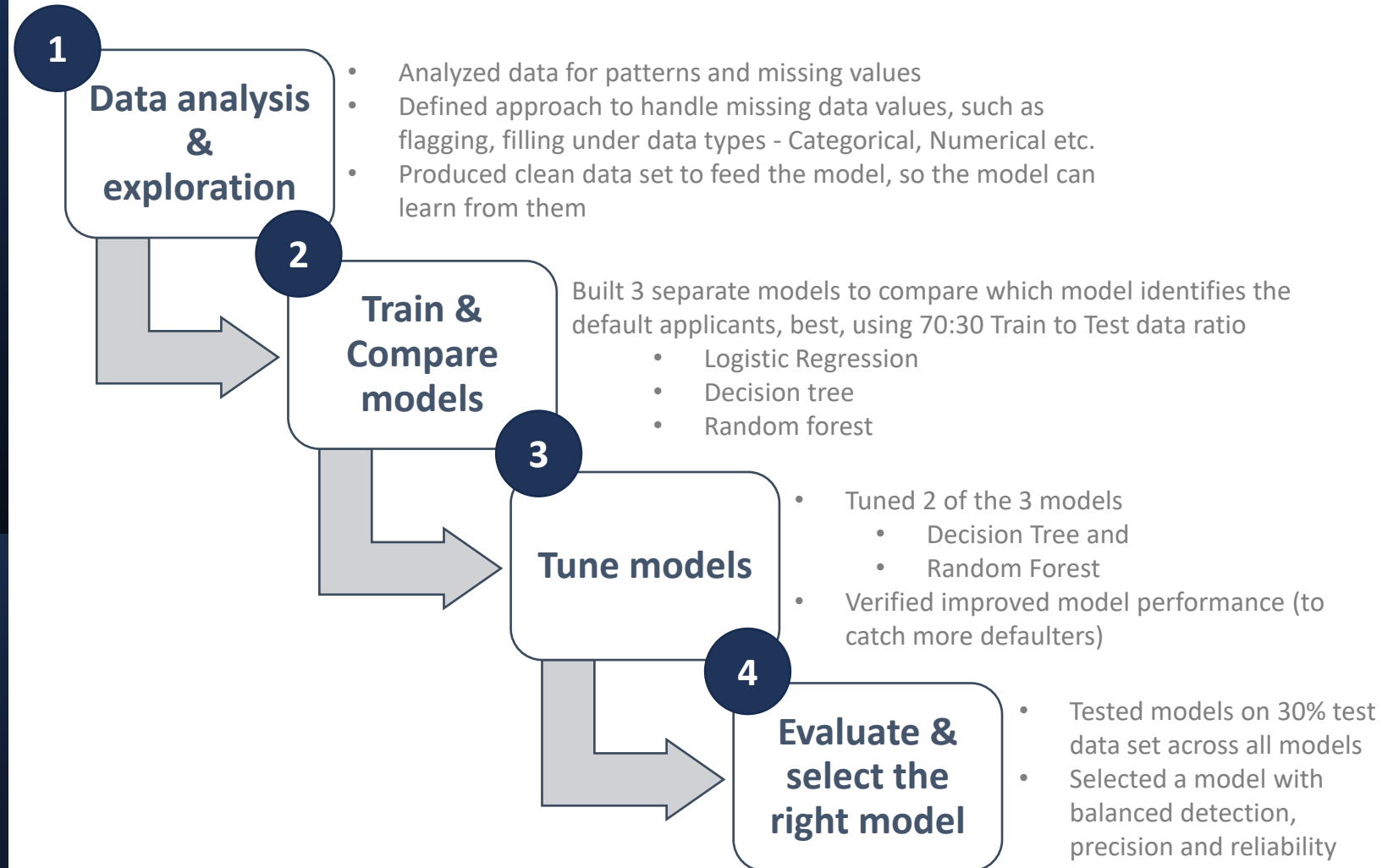
21.26%

~1 in 5 applicants had incomplete financial data on file at the time of application, signaling potential risks with missing Debt-to-Income values

The Model Approach

The Model Approach

From raw applicant data analysis to a production-ready risk model



Model Output & Model Performance

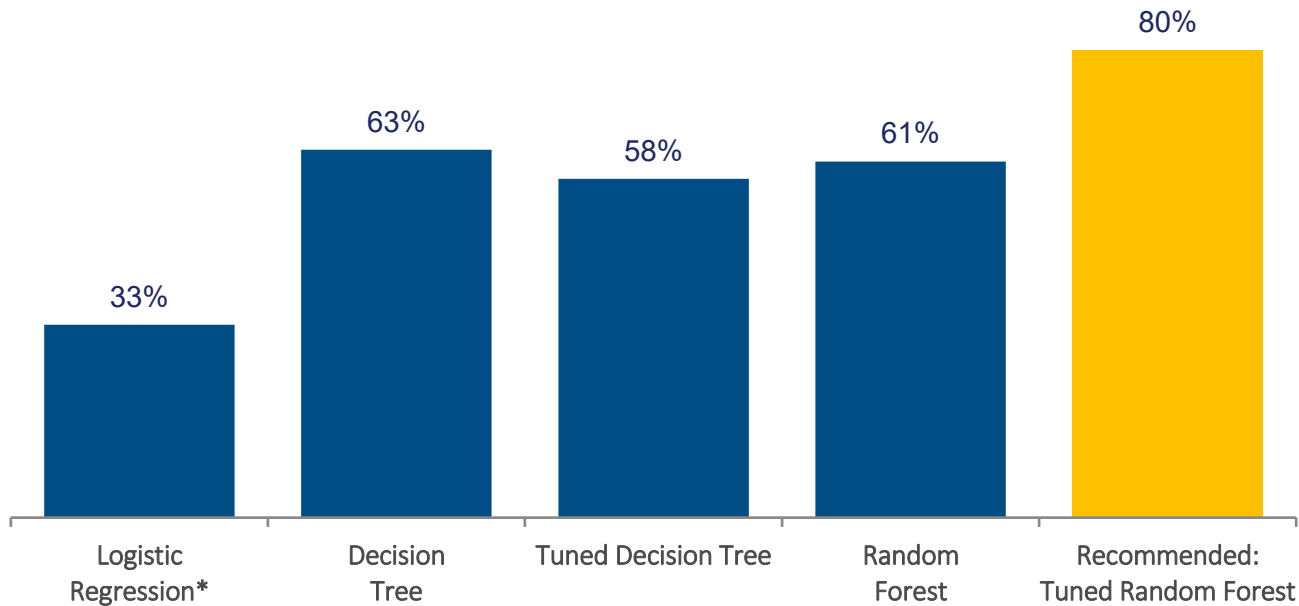
How accurately each of the model predicted the defaulters from the test data (70:30 – Training vs Test)
How long did it take to execute each model

Model	# of Defaulters	Model Prediction	% Model Prediction	Recall Score	Model execution time
Logistic Regression	399	13	3.26%	3%	5 seconds
Decision Tree	399	250	62.66%	63%	60 Seconds
Tuned Decision Tree	399	231	57.89%	58%	135 Seconds
Random Forest	399	242	60.65%	61%	45 seconds
Tuned Random Forest	399	319	79.95%	80%	320 seconds

Recall Score: measures the proportion of actual positive cases that were correctly identified by the model

Model Chart & Summary

% Defaulters accurately identified by the models using test data set



WHAT THE NUMBERS MEAN

79.9%

Catch Rate

Of every 100 applicants who defaults, the model correctly flags about 80 of them in advance.

71.0%

Flag Accuracy

Of every 100 applicants the model flags as high-risk, about 71 applicants genuinely go on to default.

88.3%

Overall Accuracy

Across all applicants—approved and declined—the model's prediction matches the actual outcome 88% of the time.

Key Findings

Key Findings: What Drives Default Risk

#	Feature	Observations from the data analysis
#1	Debt-to-income	The single strongest predictor, with the missing data by 21.26%. Defaulters have a noticeably higher median debt-to-income ratio, making it one of the most meaningful predictors of default
#2	New Credit Inquiries	Applicants with several recent credit inquiries, or a short credit history default more. Both suggesting financial stress or inexperience
#3	Job Stability	Short job tenure with variable income (Sales, Self-employed) carry meaningfully higher default rates than salaried professionals with longer tenure and increased job stability
#4	Loan Purpose	Home improvement loans default slightly more often than debt-consolidation loans
#5	Loan Size & Collateral	Loan amount and property value have a smaller, but consistent effect on risk

Recommendations

Recommendations & Next Steps

1

Deploy the Tuned
Random Forest Model

Given the model performance, deploy it on production to be used as risk-scoring layer in the underwriting process to flag high-risk applicants, for a closer review.

2

Data completeness check
during application process

Ensure data completeness during application process by imposing mandatory data requirement fields, for Credit History, Debt-to-income etc., which will close the gap on missing data and improve fairness and accuracy.

3

Update underwriting
policy

Update Underwriting policy by building explicit review triggers around debt-to-income, recent credit inquiries, short credit history and higher-risk job categories.

4

Pilot the model

Run the model in parallel with current processes for at least one quarter to validate real-world performance and calibrate the risk threshold to the bank's cost tolerance.

5

Monitor and retrain the
model

Track model accuracy against actual outcomes quarterly and retrain as borrower behavior and the economy shift.

Business Impact

Business impact upon model implementation

What benefits do the **Loan Default Prediction** model offer to the bank

+20 pts More Defaulters Caught

- Detection improves from ~60% (baseline models) to ~80%, resulting in fewer defaults slipping through the underwriting

Fewer Credit Losses

- Earlier, more accurate risk flags mean fewer bad loans are approved in the first place, directly reducing bank charge-offs

Faster Underwriting Decisions

- A consistent, automated first pass lets underwriters focus their time on the applications that truly need judgement

Sharper Lending Policy

- Clear, ranked risk factors give the bank concrete, defensible criteria to refine approval and pricing guidelines

Conclusive Recommendation: Approve a pilot rollout of the Tuned Random Forest model alongside current underwriting for one quarter

APPENDIX

Weblinks for statistics on US Home Equity Loans

US Home Equity Loans Delinquency statistics (90 days)

https://ycharts.com/indicators/us_home_equity_loans_delinquent_by_90_day

Default Probability and Loss, given historical defaults

<https://www.occ.gov/publications-and-resources/publications/economics/working-papers-archived/economic-working-paper-2014-1.html>

Current Home Equity Loan Rates

https://www.wsj.com/buyside/personal-finance/mortgage/home-equity-loan-rates?eafs_enabled=false

Weblinks for statistics on US Home Equity Loans

Nationwide delinquency rate report

<https://www.lendingtree.com/home/mortgage/u-s-mortgage-market-statistics/>

Data on Mortgage lending | NCUA

<https://ncua.gov/newsroom/press-release/2024/ffiec-publishes-2023-data-mortgage-lending>